

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network
Security COURSE CODE: CSA5116

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks:50

Annexure A

CONCEPT MAPPING

Code of Conduct:

I, VMuralekarsbitt (192312567) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Co1-At-2

1, Draw a Concept map linking number theory concepts used in cryptography:- Modular Inverse, Extended Euclid Algorithm, Fermat's little theorem, Euler's Phi function and Euler's Theorem. Perform the following tasks:-

a, show how each concept is mathematically considered

b, Map their role in encryption/decryption.

c, Analyze which concept is most critical for public key & justify.

1, Modular Inverse:

For an integer a , its modular inverse a^{-1} modulo m satisfies:-

$$a \times a^{-1} = 1 \pmod{m}$$

inverse exists when,

$$\gcd(a, m) = 1$$

Ex:- 3 modulo 7 [inverse]

$$3 \times 5 = 15$$

$$15 \pmod{7} = 1$$

$$3^{-1} \equiv 5 \pmod{7}$$

→ Modular inverses are important in deriving private/decryption operations from public parameters.

2, Extended Euclidean Algorithm:-

The Extended Euclidean Algorithm calculates:-

$$\gcd(a, b).$$

also finds integers x & y satisfying

$$ax + by = \gcd(a, b) \text{ if } \gcd(a, m) = 1.$$

Thus, the extended euclidean algorithm is directly connected to modular inverse calculation.

3, Euler's phi Function:-

→ Represented as:- $\phi(n)$

→ It gives the no. of +ve integers less than (or) equal to n that are relatively prime to n .

For two distinct primes p & q :-

$$\phi(pq) = (p-1)(q-1)$$

Ex:- $p=3, q=5$

$$n = pq = 15$$

$$\phi(15) = (3-1)(5-1)$$

$$\phi(15) = 2 \times 4 = 8.$$

→ Euler's phi function is particularly important in RSA key generation.

4, Euler's Theorem:-

→ States:- $a^{\phi(n)} \equiv 1 \pmod{n}$

$$\gcd(a, n) = 1$$

This theorem provides the mathematical foundation for many modular exponentiation operations used in public-key cryptography.

5, Fermat's Little Theorem:-

For a prime number p , if:-

$$a^{p-1} \equiv 1 \pmod{p} \quad p \nmid a.$$

This can be viewed as a special case of Euler's theorem

$$\phi(p) = p-1$$

$$a^{\phi(p)} = a^{p-1} \equiv 1 \pmod{p}$$

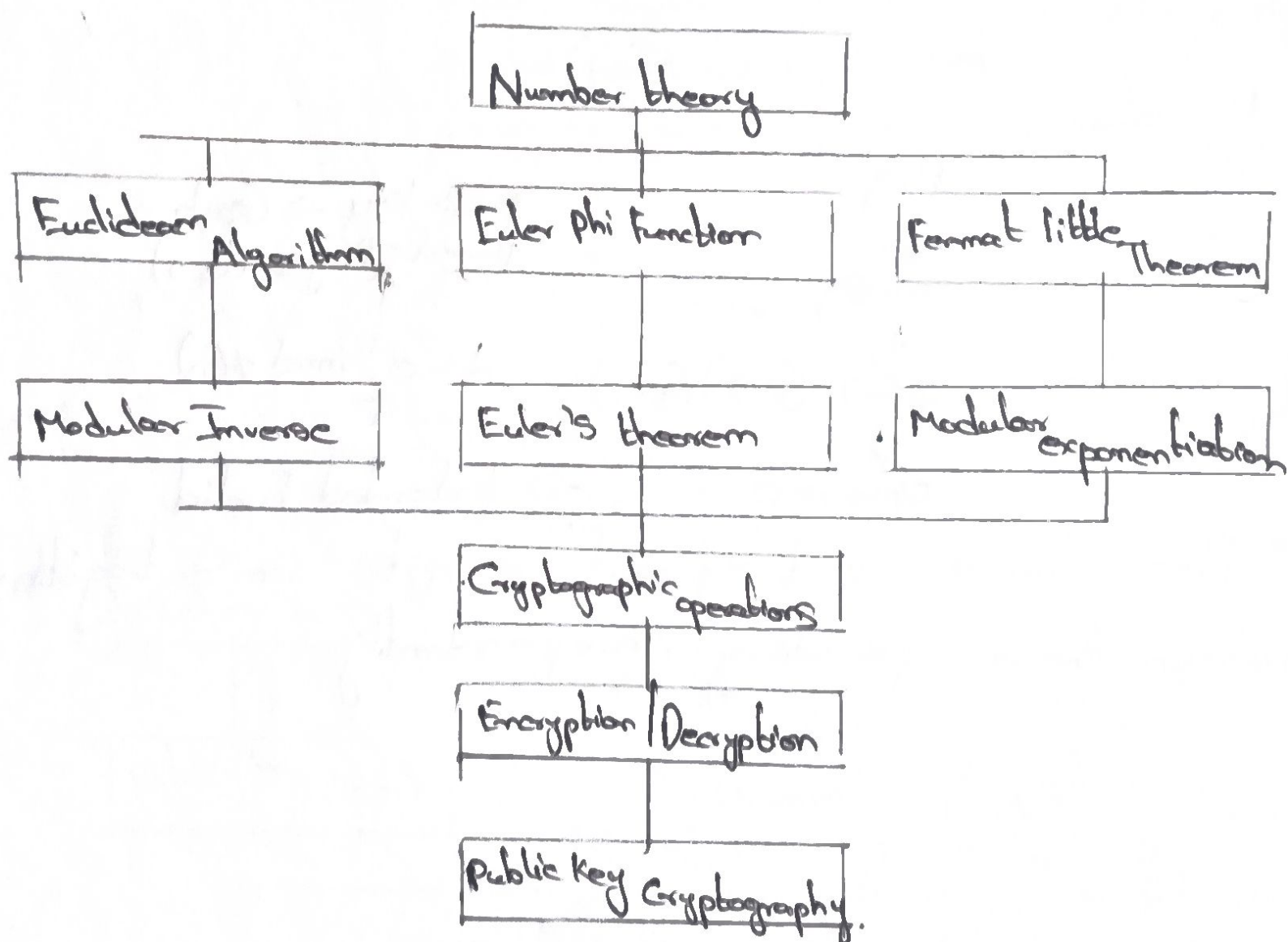
Connection:-

Euler's Theorem



Fermat's little theorem

a)



b)

Concept	Main mathematical role	Cryptographic importance
Modular inverse	Finds $a^{-1} \pmod{m}$	Helps calculate inverse operations
Extended Euclid	Computes gcd & modular inverse	Efficient inverse calculation
Euler's phi	Calculates $\phi(n)$	Important in RSA
Euler's Theorem	Establishes modular exponentiation relationship	Supports RSA correctness
Fermat's little theorem	Modular arithmetic for primes	Useful in number theoretic cryptography

C, For RSA, Euler's phi function is the most directly critical among the listed concepts for constructing the

$$\begin{aligned}n &= pq \\ \phi(n) &= (p-1)(q-1) \\ \gcd(e, \phi(n)) &= 1 \\ ed &\equiv 1 \pmod{\phi(n)}\end{aligned}$$

Therefore,

$$\begin{aligned}p, q \\ \downarrow \\ n = pq\end{aligned}$$

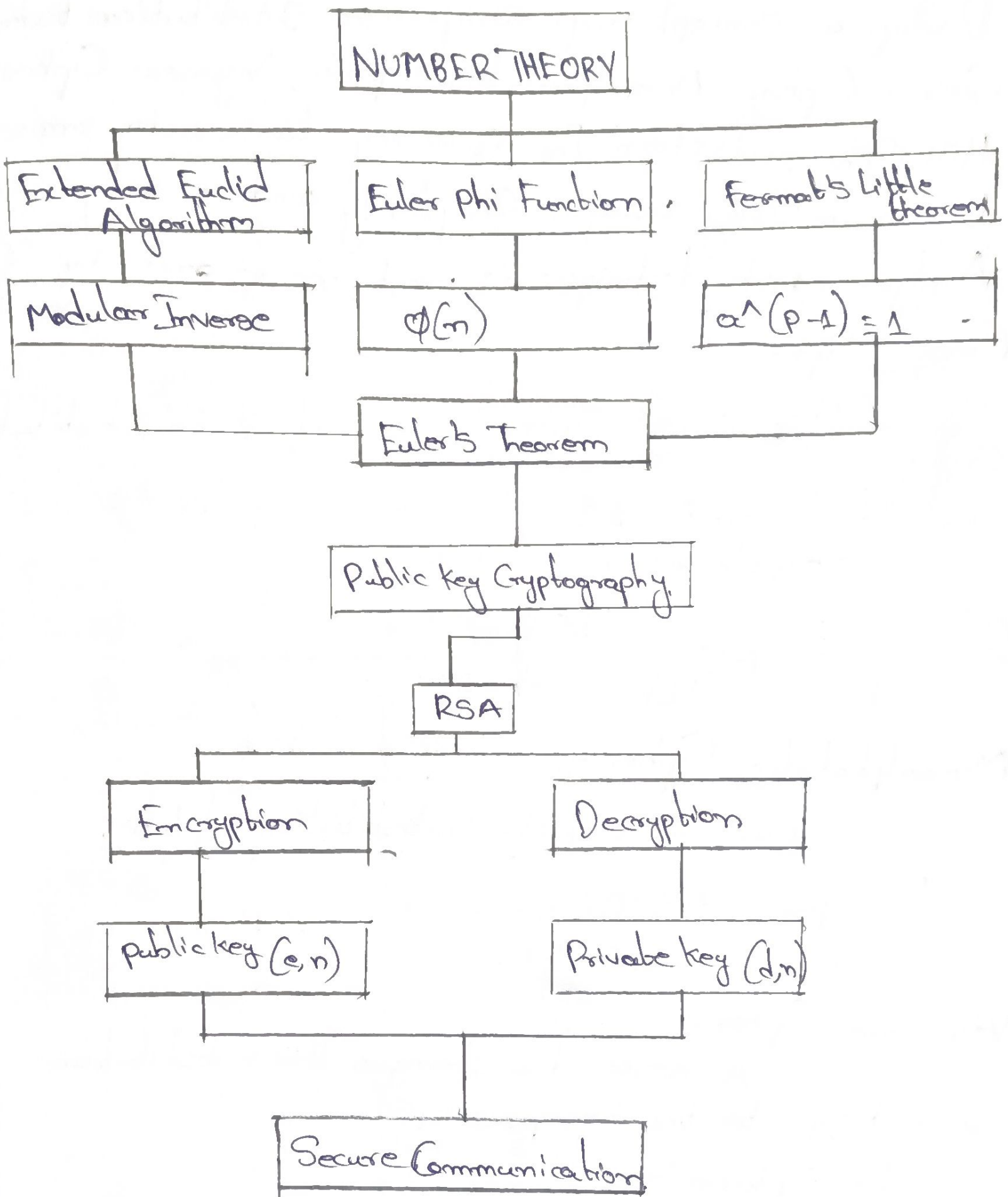
$$\begin{aligned}\downarrow \\ \phi(n) = (p-1)(q-1) \\ \downarrow \\ \text{choose } e\end{aligned}$$

$$\begin{aligned}\text{Public key} &\rightarrow (e, n) \\ \text{Private key} &\rightarrow (d, n)\end{aligned}$$

$$d = e^{-1} \pmod{\phi(n)}$$

$\rightarrow$ Extended Euclid

This shows that several concepts work together rather than operating independently



- 2) Develop a concept map comparing Substitution technique, Caesar Cipher, Monoalphabetic Cipher, Vigenere Cipher, Hill Cipher. Perform the following tasks to connect based on: key usage, Complexity and Security strength. Analyze which technique is most secure and why.

Caesar Cipher:-

The Caesar Cipher is a fixed-shift Substitution Cipher.

Ex:- Shift = 3

Plaintext:- A B C D E

Ciphertext:- D E F G H

Encryption,

$$C = (P + K) \bmod 26$$

Monoalphabetic Cipher:-

Uses a complete substitution alphabet

Ex:-

Plaintext:- A B C D E F ...

Ciphertext:- Q W E R T Y ...

Vigenere Cipher:-

A keyword & changes the substitution according to the key position.

Ex:- Plaintext:- H E L L O

Key:- K E Y K E

Encryption, $C_i = (P_i + K_i) \bmod 26$

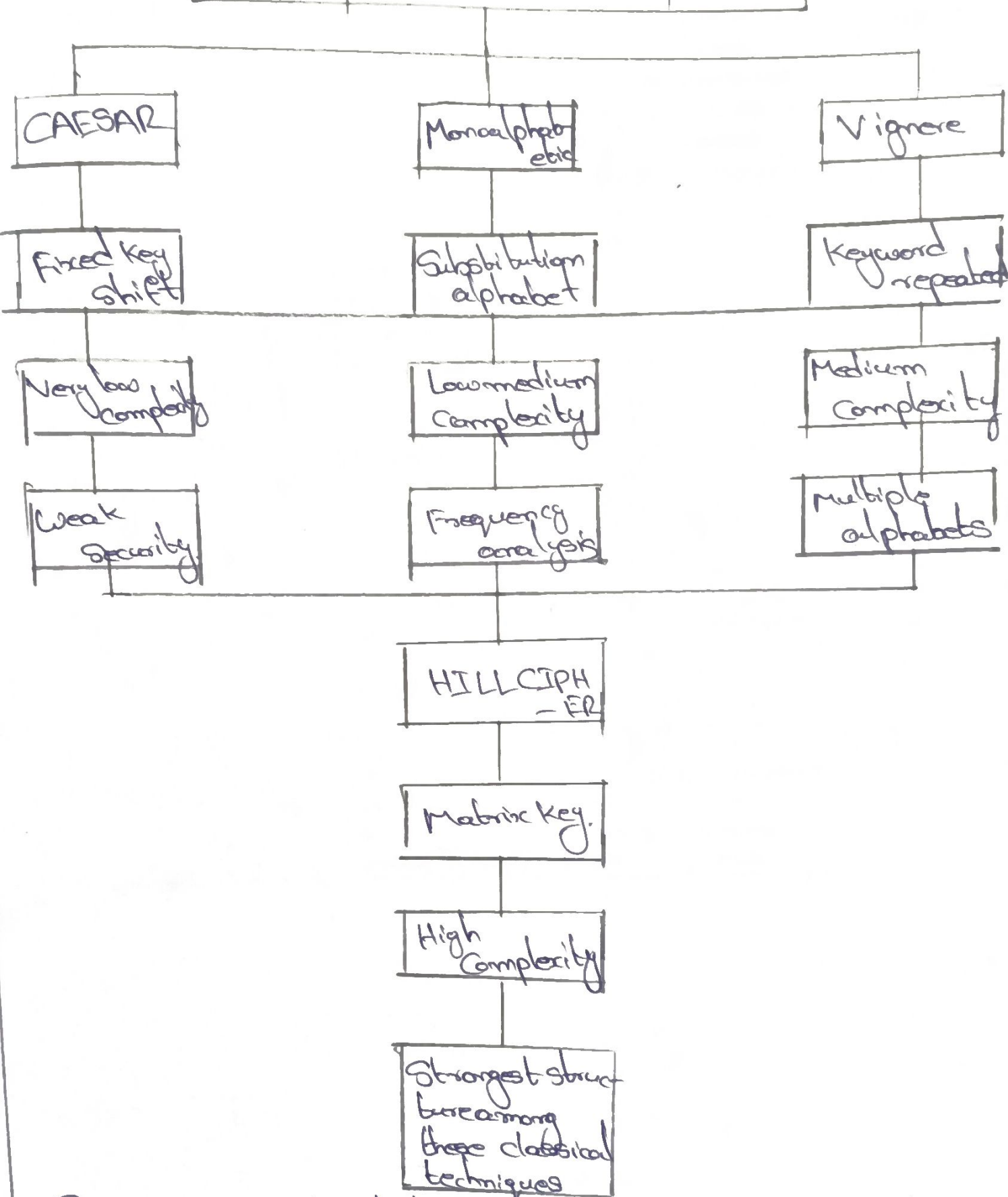
Hill Cipher:-

Uses a matrix mathematics for encryption

$$C = KP \bmod 26$$

$$P = K^{-1}C \bmod 26$$

SUBSTITUTION TECHNIQUES



Caesar < Monoalphabetic < Vignere < Hill in terms of increasing mathematical complexity, although all four are classical techniques & are unsuitable as standalone modern security algorithms.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding ; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation